

Investigating laser-induced thermal responses of C/SiC composites within airflow conditions

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HIGHLIGHTS

- Investigation of high-energy laser irradiation's impact on the thermal behavior of C/SiC composites in airflow conditions.
- A novel model for effective thermal conductivity, incorporating reaction mechanisms and by-product considerations in C/SiC composites.
- Detailed analysis of temperature variations and ablation depths in C/SiC composites under different laser power densities.
- Assessment of the influence of airflow on the thermal ablation and temperature profiles of laser-irradiated C/SiC composites.

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ABSTRACT

This study primarily investigates the thermal effects of high-energy laser irradiation on carbon fiber-reinforced silicon carbide (C/SiC) composites in an airflow environment. Experiments conducted within a wind tunnel at various power densities elucidate the impact of laser ablation on the material's surface and rear temperatures, ablation depth, and morphology. The research highlights the intriguing thermal effects of high-energy laser on C/SiC materials under airflow conditions and introduces a novel effective thermal conductivity model that incorporates reaction mechanisms and by-product considerations. This model significantly enhances the accuracy in predicting temperature variations and ablation rates, offering critical insights into the thermal behavior of C/SiC composites under extreme conditions.

1. Introduction

The burgeoning interest in exploring extraterrestrial destinations such as Mars, Venus, the gas giants, and the icy giants, alongside the increasing frequency of missions returning to Earth, underscores a pressing need for advanced thermal protection systems (TPS) to safeguard future space missions [1]. In this context, carbon fiber-reinforced silicon carbide (C/SiC) composites have garnered attention as a promising material due to their outstanding mechanical strength, high-temperature resilience, and superior impact resistance [2]. Recent studies, such as Nutakki et al. [3] and Faisal et al. [4], emphasize the increasing importance of heat transfer research. The investigation into their thermal conductivity is crucial for optimizing the resilience and performance of aerospace structures [5–7]. Recent advancements in high-power laser ablation studies have exploited the unique properties of lasers, such as high spectral purity and temporal coherence, to simulate extreme thermal shock and stress conditions on these materials [8].

Pertinent to the study of ablation behavior, Wang et al. [9] explored the ablation dynamics of 2-D C/SiC composites under continuous laser exposure in both tangential supersonic airflow and static air settings, elucidating the effects of airflow on ablation and the zonal behaviors within the ablation region. Complementary research by Pan et al. [10] investigated the impact of laser

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parameters on the ablation mechanisms of C/SiC composites, highlighting significant factors such as power density in modifying surface morphology and ablation behavior. Additionally, Wang et al. [11] identified an “avalanche” phenomenon during laser irradiation of C/SiC composites in a hypersonic wind tunnel, observing an increased ablation rate under tangential hypersonic airflow compared to static conditions. Wang et al. [12] further clarified the ablation behavior under hypersonic airflow by interpreting the forward movement of ablation using wall shear stress. Moreover, recent studies have provided deeper insights into the multi-scale ablation behavior of C/SiC composites. Wang et al. [13] conducted ablation experiments and multi-scale numerical simulations, revealing how different power densities affect surface temperature and ablation morphology, and established a macro-mesoscopic cross-scale ablation model incorporating solid heat transfer, chemical reactions, and temperature gradients between fibers and matrix. Another study by Wang et al. [14] used a CW laser to simulate extreme thermal environments, examining the coupling effects of high-temperature airflow and localized heat sources on the ablation behavior of C/SiC composites, and developed a multi-field coupling numerical model to capture fluid–solid–thermal–ablation interactions. Additionally, research by Wang et al. [15] investigated the ablation behavior under local structural thermal sources, demonstrating significant differences in ablation pit morphology under various heat flux densities and predicting ablation behavior using improved numerical methods.

These studies primarily address the influence of high-speed airflow on ablation rates. However, their simulation processes often neglect to account for the material temperatures during ablation, and the thermal conductivity of materials is frequently assumed to be a fixed value. Such oversights can introduce considerable inaccuracies into ablation research. Therefore, in order to advance our understanding of ablation rates and morphologies under airflow conditions, it is imperative to carefully consider the impact of thermal conductivity, which serves as a pivotal parameter in heat transfer processes. Abbas et al. [16] investigated the thermal performance of a hybrid nanofluid consisting of a suspension of aluminum alloys (AA7075-AA7072) in a methanol-base fluid, highlighting that increasing the concentration of nanoparticles enhances the thermal conductivity and improves the thermal efficiency of the working fluid. Liu et al. [17] assessed the effective thermal conductivities of plain-woven C/SiC composites, integrating in-yarn and out-of-yarn porosities via geometric reconstruction models to evaluate the thermal conductivity of the Representative Volume Element (RVE) while accounting for porosity. Additionally, Abbas et al. [18] analyzed heat and mass transfer in the magnetohydrodynamic boundary layer flow of a second-grade nanofluid over an inclined stretching permeable surface within a porous medium, underscoring the impact of a porous medium on heat transfer. Research by Li et al. [19], employing the Finite Element Method (FEM) for thermal analysis of glass-fiber reinforced plain-woven composites, examined the influence of fiber volume fraction on in-plane and out-of-plane thermal conductivities at the microscale. Furthermore, Gasmi et al. [20] examined the thermal performance of a motile-microorganism within two-phase nanofluid flow for various non-Newtonian models on static and moving surfaces, considering the effects of heat generation, chemical reaction, and activation energy. The effect of Pyrolytic Carbon (PyC) was found to be minimal in multiscale numerical modeling of the thermal behavior of plain-woven composites with interfacial and internal defects [21]. Investigations into the impact of weaving structure, utilizing FEM to ascertain Effective Thermal Conductivities (ETCs) for different braided structures, indicated minimal variance in ETCs for identical braiding techniques (2D, 2.5D, 3D) with varying structures [22], acknowledging interface effects and emphasizing the challenge of achieving comprehensive interfacial contact among different components, leading to temperature discontinuities [23].

Existing research [24] primarily focuses on temperature ranges below 1000 °C. However, specialized temperature changes on the ablation surface during experiments have not been adequately addressed. Furthermore, the temperature conditions on the backside of materials, known to affect aircraft operation, have been largely overlooked. This paper aims to investigate temperature variations up to 3000 K by conducting laser irradiation experiments at various power levels, encompassing both surface and backside temperatures. By establishing an effective thermal conductivity model that incorporates reaction mechanisms, the paper elucidates some novel experimental phenomena observed under different laser power densities, achieving better temperature fitting compared to existing methods. This provides insights into the thermal behavior of C/SiC composite materials under extreme conditions.

2. Experimental setup

2.1. Specimen preparation

The 2.5D-C/SiC composite plates investigated in this study were synthesized at the State Key Laboratory of Ultrahigh Temperature Composites, Northwestern Polytechnical University. The fabrication process commenced with the weaving of T300 carbon fiber tows into 0°/90° carbon cloths, followed by the stacking of twelve layers in the thickness direction to form the preform structure. Subsequently, the SiC matrix was formed via a six-cycle chemical vapor infiltration (CVI) process, conducted at temperatures ranging between 950 °C and 1200 °C under a pressure setting of 5000 Pa. Pyrolytic carbon (PyC) interlayers, each 200 nm thick, were deposited using C₃H₆ at 850 °C and 5000 Pa. Following these steps, additional C₃H₆ deposition was carried out on the C/SiC semi-finished products to enhance their properties before shaping them into the required specimens. Details regarding dimensions and specific attributes are provided in Table 1, including thermal conductivity (k), coefficient of thermal expansion (CTE), and thermal diffusivity (TD). These values were measured by the State Key Laboratory of Ultrahigh Temperature Composites, Northwestern Polytechnical University, which ensures their accuracy and reliability. It is worth noting that the inherent limitations of the CVI process result in some degree of porosity within the composites, thereby preventing complete internal densification. The volume fraction of carbon fiber is approximately 40%, with a porosity of around 6%. C/SiC samples are manufactured into flat plates measuring 50 × 50 mm in length and width, with a thickness of 5 mm.

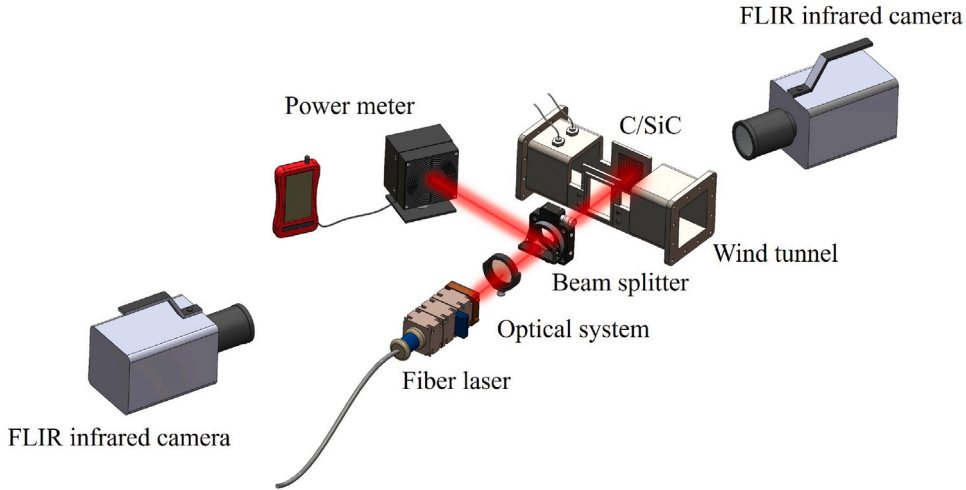


Fig. 1. Experimental setup for laser ablation in the wind tunnel.

Table 1

Typical properties of C/SiC specimen in this experiment.

Parameter	1	2	3	4	5	6	7	8	9
T (°C)	20	100	200	300	400	500	600	700	800
k (W/m K)	7.67	8.20	7.89	7.64	7.36	7.16	7.17	7.01	6.89
T (°C)	300	600	900	1000	1100	1200	1300	1390	
CTE (10 ⁻⁶ /K)	3.92	5.03	5.39	5.91	5.96	6.21	6.95	7.81	
T (°C)	328	496	871	1022	1125	1276	1398		
TD (10 ⁻² cm ² /s)	0.034	0.03	0.025	0.024	0.024	0.023	0.023		

Table 2

Modular Optical Profilometer ST400 -NANOVEA.

Scan options	Dimension	Position	Scanning type	Pen
Parameter	Area	Starting	Single direction	10 mm

Table 3

Laser parameters.

Setting power (W)	Average power (W)	Spot diameter (mm)	Effective density (W/cm ²)	Duration (s)
3300	2680	24.5	330	30
6200	5630	24.6	650	30
13 500	12 300	24.5	1420	20

2.2. Experimental setup

The experimental configuration, depicted in Fig. 1, utilized an IPG YLS fiber laser with a wavelength of 1070 nm to conduct ablation studies on C/SiC composites. To counteract the instability associated with high-power laser output, a series of lenses were employed to refine the optical path and enhance beam quality. The diagram simplifies the setup to an optical system, incorporating a beam splitter and a high-power power meter for output power calibration. These experiments were conducted in a free jet wind tunnel at the State Key Laboratory of Laser Interaction with Matter, with wind speed meticulously maintained at 0.6 Ma via system software, ensuring consistent experimental conditions. The laser was aligned perpendicularly to the surface of the C/SiC specimen through a transparent viewing window. Temperature data for both the front and back surfaces were acquired using colorimetric temperature measurement techniques, with a FLIR infrared camera facilitating the data collection. To comprehensively analyze the ablation morphology, we employed the Modular Optical Profilometer ST400-NANOVEA, as shown in Fig. 2, with the specific parameters detailed in Table 2. This analytical step is crucial for understanding the alterations on the C/SiC surface post-ablation, providing insights into how different laser power settings influence the material. Given the significant influence of laser power on the ablation and temperature behavior of C/SiC composites, a calibration process for the laser output power and the actual laser spot measurement was necessary to address potential errors arising from high laser power instability. For precise determination of the laser output power, a power meter was used, while the laser's spatial mode was evaluated using an LT665 Beam Profiling Camera (OPHIR), with detailed laser parameters listed in Table 3.

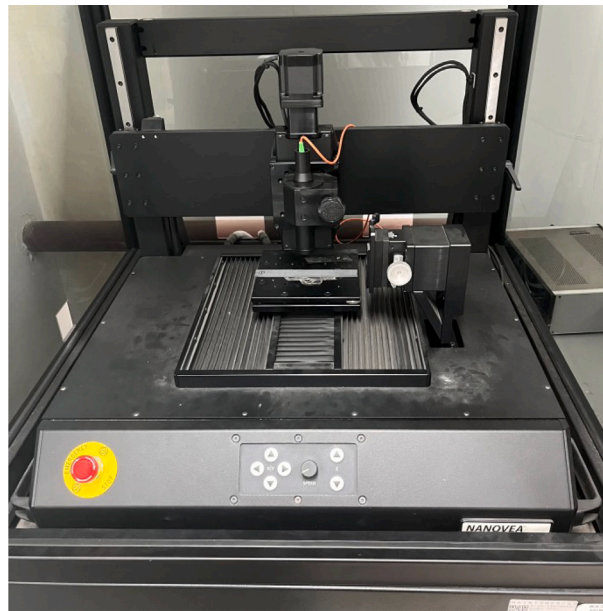


Fig. 2. ST400-NANOVEA.

3. Results and discussion

3.1. Thermal responses

Ablation experiments were conducted at three distinct power densities, specifically set at 0.6 Ma, to elucidate the thermal response of the C/SiC composites. The temperature variations on the surface, as captured in Fig. 3, refer to the maximum surface temperatures recorded during each experimental setup.

In the initial experiment (Test1), the maximum temperature recorded on the front side was 1602 °C, while the peak temperature on the back side reached 382 °C, resulting in a temperature differential of 1220 °C. Throughout the 30-second laser irradiation period, the temperature exhibited a trend towards stabilization.

The second experiment (Test2) revealed a peak temperature of 2365 °C on the front side and a minimum of 772 °C on the back side, with significant temperature fluctuations observed beyond 1700 °C. This indicates a correlation between the increase in laser power and both the maximum temperature and temperature gradient. Moreover, noticeable temperature jitter was observed during the ablation process, with a marked deceleration in the temperature rise rate beyond 1700 °C.

The third experiment (Test3) exhibited a rapid escalation in temperature, reaching a maximum of 3032 °C. Following this peak, the temperature showed a progressive decline, even under continuous laser exposure, an effect attributed to the enhanced heat dissipation capacity resulting from C/SiC structural alterations post rapid ablation at temperatures exceeding 3000 °C.

Additionally, examining the temperature variation curves at different ablation powers reveals that, at power densities below 650 W/cm², the temperature differential between the front and back surfaces of the material increases. Moreover, the temperature rise rate on the material's back surface increases with the rise in power density, with a faster temperature increase rate observed at higher power densities.

A more in-depth analysis reveals several key insights:

For Test1 at 330 W/cm², the temperature stabilization indicates that the formation of a protective oxide layer (SiO₂) helps to mitigate further temperature rise, suggesting effective thermal insulation properties of the oxide layer.

In Test2 at 650 W/cm², the significant fluctuations beyond 1700 °C suggest the onset of phase changes and rapid oxidation reactions, contributing to thermal instability. The observed jitter in temperature rise can be attributed to the dynamic formation and removal of oxide layers, impacting the heat transfer mechanisms. These fluctuations indicate that not all liquefied products are entirely blown away from the ablation area, affecting the overall thermal behavior of the material.

Test3 at 1420 W/cm² shows a peak temperature followed by a decline, indicating rapid ablation and material removal at extremely high temperatures. This phenomenon can be explained by the increased heat dissipation due to structural changes in C/SiC, leading to a faster cooling rate on the surface. The significant removal of material due to ablation results in a reduction in material thickness. According to the heat transfer equation, as the thickness decreases, the heat conduction from the surface to the rear becomes more efficient, resulting in faster heat dissipation on the surface and a temperature drop, while the rear temperature rises more rapidly. This effect is less pronounced at lower power densities due to slower ablation rates but becomes evident at higher power densities. Additionally, because the temperature collector on the rear surface has a limited range, a flat region appears in the temperature readings during the later stages of ablation when the temperature exceeds this range.

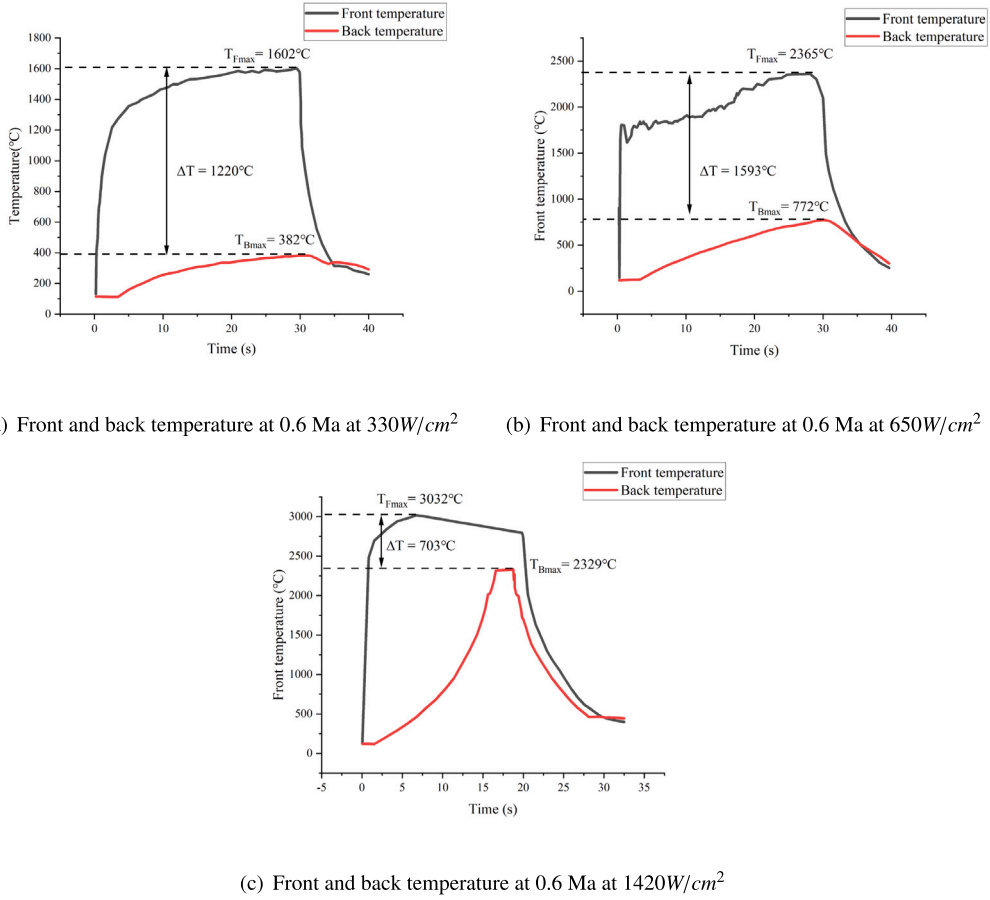


Fig. 3. Temperature variation curves of material front and back surfaces at different power densities.

3.2. Ablation morphology

The macroscopic examination of the ablated material provides an intuitive understanding of the alterations induced by laser exposure. Fig. 4 showcases the macroscopic morphology of the irradiated C/SiC composites under various laser parameters. The first column delineates the macroscopic appearance post-irradiation, the second column represents a schematic of the ablation region's segmentation, and the third column provides a schematic diagram of the ablation zone, using ST400-NANOVEA for ablation depth detection mapping.

Combined with the surface temperature data from Fig. 3, the primary SiC reactions during laser ablation are inferred as follows [25,26]:



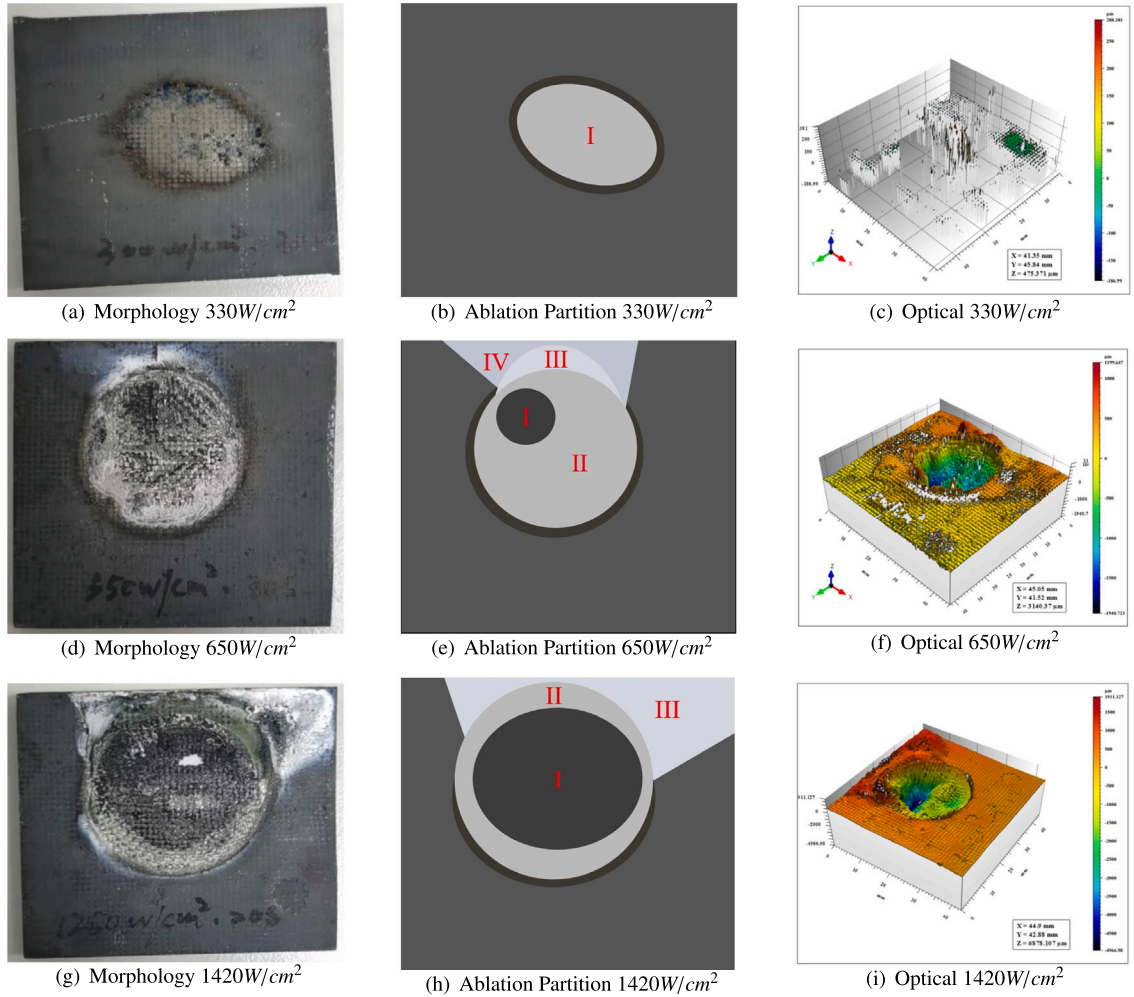


Fig. 4. Comparison of 0.6 Ma ablation morphology.



At the initial stage of irradiation, within the temperature range of 700°C – 1723°C , and under a laser power density of 330 W/cm^2 , the primary reactions involve the transformation of SiC into SiO_2 and carbon oxides, as represented in reactions (1)–(3). In this context, Region I illustrates the formation of a white solid SiO_2 layer, surrounded by a blackened area indicative of post-ablation SiC. The maximum ablation depth measured was 0.186 mm , correlating with the peak temperature of 1602°C , suggesting minimal ablation under stable oxide protection.

Increasing the laser power density to 650 W/cm^2 raises the surface temperature to 2365°C , initiating reactions (4) and (5) where solid SiO_2 transitions into its liquid form, forming Region II. The airflow at 0.6 Ma assists in dispersing high-temperature oxides in Region I, while Region III witnesses the re-solidification of SiO_2 . Under these conditions, both solid SiO_2 ($\text{SiO}_2(s)$) and transitioning SiO_2 ($\text{SiO}_2(l)$) contribute to temperature fluctuations and gradual temperature rise, as not all liquefied products are entirely blown away from the ablation area. Ablation morphological analysis delineates distinct zones: bare SiC (I), non-liquefied SiO_2 (II), partially removed SiO_2 (III), and vaporized deposition areas (IV), with a maximum ablation depth of 1.94 mm .

At a higher laser power of 1420 W/cm^2 , the surface temperature exceeds 3032°C , predominantly driving reactions categorized under (6)–(9). Accompanied by the rapid removal of a significant amount of liquefied products by airflow, when the temperature quickly exceeds 2830°C , the products vaporize. This phase is characterized by a larger exposed C/SiC area (I) compared to previous stages. Due to the Gaussian distribution of laser power density, there are solid SiO_2 ($\text{SiO}_2(s)$) (II) in the surroundings, and during rapid temperature rise, both liquid SiO_2 ($\text{SiO}_2(l)$) and gaseous SiO_2 ($\text{SiO}_2(g)$) are transported to downstream regions (III). Ablation occurs around 17 s , resulting in material perforation.

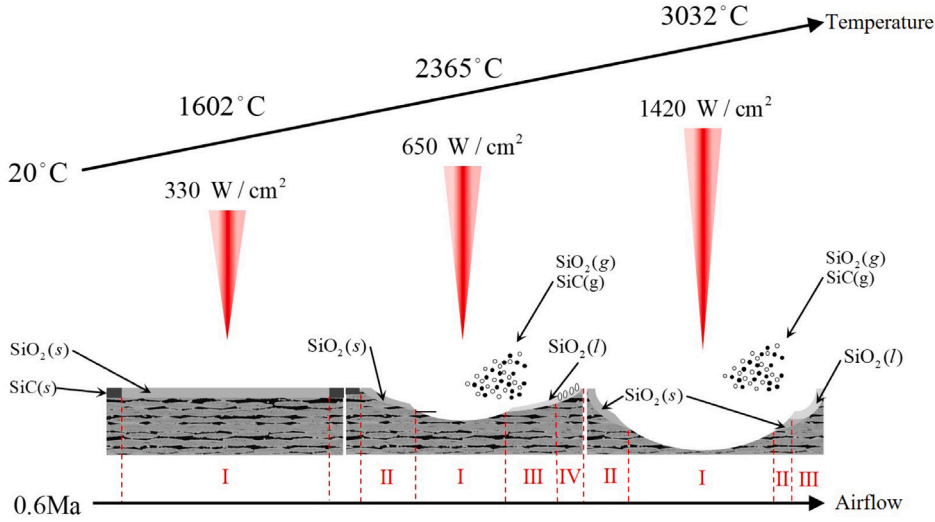


Fig. 5. Ablation mechanism.

3.3. Numerical method and verification

3.3.1. Thermal conductivity analysis of C/SiC

To elucidate the unique phenomena observed during the experimental process, such as the significant temperature difference between the irradiated and the rear side of the material during laser irradiation, the temperature fluctuations and gradual rise on the material's surface at specific ablation temperatures, and the peculiar rate of temperature increase on the irradiated backside, we delved into the effective thermal conductivity within the heat transfer process. By correlating the temperature curves obtained from the experiments with the ablation products in different regions, we crafted an ablation mechanism analysis diagram that evolves with the laser irradiation power density. Fig. 5 delineates the three characteristic ablation stages, accentuating the dynamic morphological transitions under various laser conditions. Integrating the information from the figure, we categorized the ablation into three distinct temperature phases. Based on the surface products and microstructures at different temperature stages, we developed a temperature-based model to calculate the effective thermal conductivity, basing our model on a comprehensive analysis of the observed ablation behavior and temperature trends.

For the heat transfer analysis that includes an oxide layer, we employed a series heat transfer model. The expression for effective thermal conductivity, k_{eff} , at elevated temperatures is provided by:

$$k_{eff} = \frac{R_{total}}{L_{total} \times A} \quad (10)$$

where R_{total} denotes the total thermal resistance of the composite layers, L_{total} represents the material's thickness, and A signifies the cross-sectional area related to heat flow.

Thermal resistance analysis encompasses the contributions from SiO_2 , C/SiC, and the contact resistance between the oxide layer and the composite, expressed as:

$$R_{total} = R_{\text{SiO}_2} + R_{\text{C/SiC}} + R_C \quad (11)$$

where R_{SiO_2} , $R_{\text{C/SiC}}$, and R_C correspond to the thermal resistances of SiO_2 , C/SiC, and the interface between the oxide layer and the composite, respectively.

Below 700 °C, C/SiC materials exhibit relative stability. The effective thermal conductivity of the material is related to several microscale factors: porosity, contact thermal resistance at interfaces, the weaving pattern of the material, and the proportion of different components in the composite material. We derived the effective thermal conductivity values from Zhang et al. [8] using a linear fitting approach. The resulting effective thermal conductivity is represented by:

$$k_{eff} = 8.0381 + 1.0 \exp(-1.0T) - 0.0015T \text{ (W/m K)} \quad (12)$$

In the temperature range of 700–1723 °C, the material undergoes oxidation reactions (1)–(3), predominantly forming $\text{SiO}_2(s)$ on the ablation surface. Additionally, a minor amount of liquid SiO_2 is also present on the ablation surface. The morphology of the oxide film formed on the ablation surface is illustrated in Fig. 6.

Guided by Zhu et al. [28], we acknowledge that SiO_2 's thermal resistance is affected by temperature, film thickness, and porosity. Thermal conductivity is determined by:

$$k_{\text{SiO}_2} = k_b \frac{pk_a + (1-p)k_b}{k_b} [1 - p^x] + k_a \frac{k_b}{pk_b + (1-p)k_a} p^x \quad (13)$$

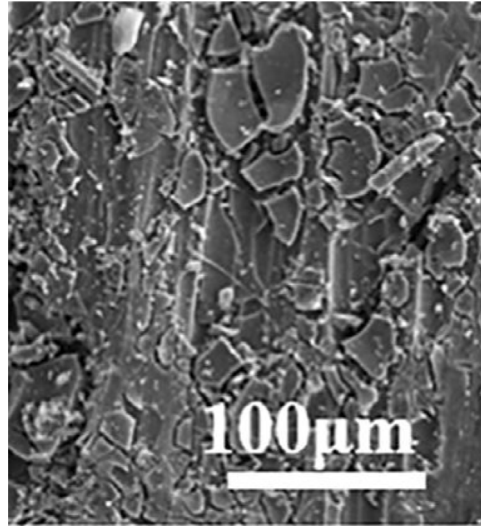


Fig. 6. Surface ablation products [27].

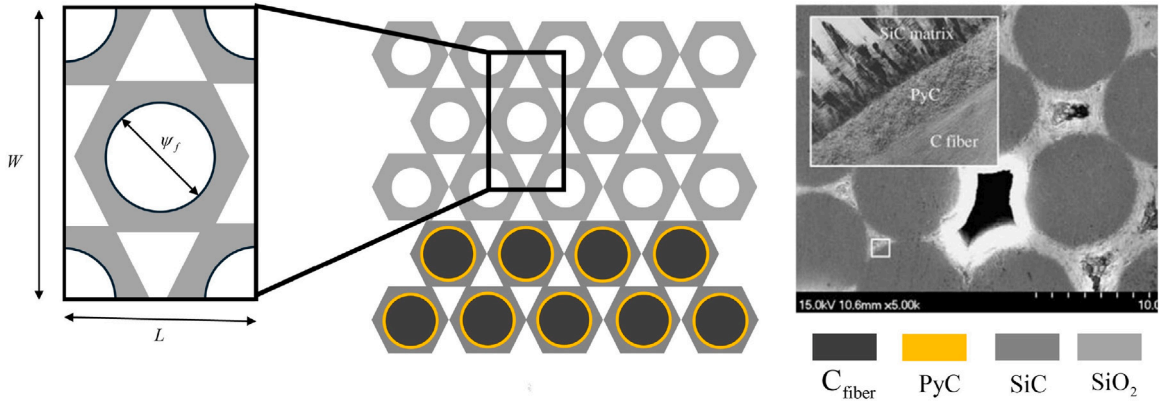


Fig. 7. Schematic representation of microscopic ablation.

In the formula, p denotes porosity, with k_a and k_b representing the thermal conductivities of the solid and porous media, respectively. The variable x serves as a fitting parameter. In the context of the Parallel (DF model) and Serial (DP model) frameworks, the weighting factors are defined as $1 - px$ and px , respectively. Typically, in this scenario, the value of x is set to 0.2.

The initial step involves determining the porosity value, followed by an analysis of the formation process of the oxide layer. Microscopically, carbon fibers adjacent to the ablation zone undergo a transformation, leading to carbide formation. Concurrently, the SiC encasing these fibers reacts, producing SiO_2 and forming a protective layer. The interplay between void defects and the layer's microstructure is pivotal in ascertaining porosity, a crucial element for computing the effective thermal conductivity of SiO_2 within the designated temperature range. For porosity assessment, we adopted a microscopic model inspired by Xu et al. [24] and illustrated in Fig. 7. This model abstracts a central geometric structure from the right-hand scheme, where the scale on the right is $10 \mu\text{m}$. The middle portion of Fig. 7 represents the post-reaction formation of SiO_2 , culminating in a summarized microscopic structure on the left. Parameter values were referenced from Xu et al. [24].

Based on the structure illustrated in the figure, we postulate that the porosity of the formed SiO_2 can be calculated using the following formula:

$$p = 1 - \frac{\pi \varphi_f^2}{2WL} \quad (14)$$

Additionally, we account for the impact of film thickness on the effective thermal conductivity, employing formulas (10) to (14). By considering the actual conditions, we calculate the material's thermal conductivity at various temperatures for thicknesses of 0.1 mm and 0.01 mm. Based on the fitting results with respect to temperature, we ultimately select a thickness of 0.1 mm as our calculation parameter.

When temperatures exceed 1723 °C, surface oxidation results in the formation of $\text{SiO}_2(l)$. Generally, the thermal conductivity of $\text{SiO}_2(l)$ is lower than that of $\text{SiO}_2(s)$. At this point, the porosity approaches zero. However, under airflow, some $\text{SiO}_2(l)$ is carried to the downstream region of the flow, which thins the oxide layer. Therefore, it is essential to introduce a velocity-related correction factor $\alpha(v)$ into Eq. (13) when temperatures are below 2230 °C. The aim is to adjust for the effects of thickness reduction and liquefaction on the effective thermal conductivity. When the temperature surpasses 2230 °C, the primary processes are reactions (6)–(9), which are chiefly concerned with oxidation or vaporization at the ablation center. At this locale, carbonization is the predominant reaction, leading to the creation of needle-like structures. During this phase, the oxide film's effect on the material's thermal characteristics is minimal and can essentially be overlooked.

The formula for interlayer thermal resistance is generally articulated as:

$$h_{co} = h_c + h_g \quad (15)$$

$$h_c = 1.25k_s \frac{m}{\sigma} \left(\frac{P}{H} \right)^{0.95} \quad (16)$$

$$h_g = \frac{k_g}{Y + M} \quad (17)$$

Here, P denotes the contact pressure, and H represents the microhardness of the softer material. In the absence of specific data, we assume the σ/m ratio, indicative of surface smoothness, ranges between 5 to 9 μm . The term h_g is the thermal resistance of the gas gap, with k_g representing the thermal conductivity of the fluid within the gap. Additionally, Y and M account for the mean plane distance and the effects of gas rarification, respectively.

Considering the negligible contact pressure P between the oxide film and the substrate, the interlayer thermal resistance is deemed minimal in this scenario.

Thus, the comprehensive thermal conductivity of the material, through its various stages, can be encapsulated as follows:

$$k_{\text{eff}} = 8.0381 + 1.0 \exp(-1.0 \cdot T) - 0.0015 \cdot T \quad (\text{W/m K}), \quad T < 700^\circ\text{C} \quad (18)$$

$$k_{\text{eff}} = \frac{L_{\text{SiO}_2} + L_{\text{C/SiC}}}{\left(\frac{L_{\text{SiO}_2}}{k_{\text{SiO}_2} \cdot A} + \frac{L_{\text{C/SiC}}}{k_{\text{C/SiC}} \cdot A} \right) \cdot A}, \quad 700^\circ\text{C} \leq T < 1723^\circ\text{C} \quad (19)$$

$$k_{\text{eff}} = \frac{L_{\text{SiO}_2} + L_{\text{C/SiC}}}{\left(\frac{L_{\text{SiO}_2}}{\alpha \cdot k_{\text{SiO}_2} \cdot A} + \frac{L_{\text{C/SiC}}}{k_{\text{C/SiC}} \cdot A} \right) \cdot A}, \quad 1723^\circ\text{C} \leq T < 2230^\circ\text{C} \quad (20)$$

$$k_{\text{eff}} = 8.0381 + 1.0 \exp(-1.0 \cdot T) - 0.0015 \cdot T \quad (\text{W/m K}), \quad T \geq 2230^\circ\text{C} \quad (21)$$

3.3.2. Coupled heat transfer model

The numerical model developed in this study tackles the transient heat transfer challenges associated with phase changes, specifically focusing on the ablation and decomposition of C/SiC composite materials. This model seamlessly integrates both physical and chemical processes, which are crucial for accurately simulating the interactions between high-velocity airflow and high-energy laser irradiation, employing a loosely coupled strategy.

We conducted a numerical simulation to analyze the ablation behavior of C/SiC composites under the dual influences of high-velocity airflow and high-energy laser irradiation. A synergistic methodology was adopted, utilizing Abaqus for structural simulations, Fluent for fluid dynamics, and MpCCI for coupling these domains, thus forming a comprehensive heat transfer model within an airflow environment. The coupling of the fluid and solid domains incorporates both aerodynamic heating and heat conduction, coordinated through a user-defined interface. To ensure both stability and precision in our simulations, we meticulously selected the time step sizes for the fluid (Δt_f) and solid (Δt_s) fields, with finer sub-step sizes applied in the solid domain to address its computational intricacies.

The fluid–thermal–ablation coupling process is depicted in Fig. 8.

The specific steps involved in the simulation process are as follows:

1. At time t_i , the wall temperature and position parameters of the C/SiC model are transferred to the flow field to establish boundary conditions for computing the flow dynamics.
2. Utilizing these boundary conditions, the flow field is recalculated to ascertain the wall heat flux, local pressure, and wall shear stress at time $t_{i+1} = t_i + \Delta t_f$.
3. These computed values of the wall heat flux are then communicated to the solid domain as boundary conditions through the MpCCI data exchange technique.
4. The input laser power density is adjusted via a subroutine, and the resulting boundary conditions facilitate the prediction of temperature distribution and material ablation at time $t_i = t_i + \Delta t_s$.

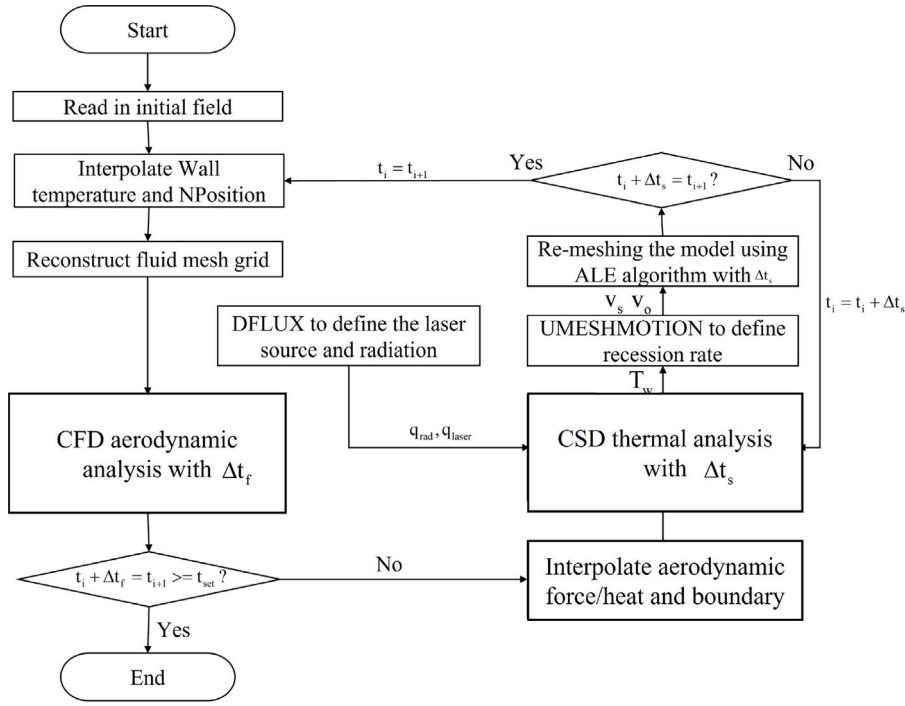


Fig. 8. Illustration of the coupling strategy.

5. Upon reaching $t_i + \Delta t_s = t_{i+1}$, the updated wall temperature and position parameters are cycled back to the fluid domain, thus serving as new boundary conditions for the flow field.
6. This iterative process (steps 1–5) continues until the completion of a coupling time step.

Given the model's inherent symmetry, we configured the mesh as shown in Fig. 10, categorizing the simulation space into fluid and solid sectors. The fluid sector, dimensions set at 75 mm × 50 mm, was subjected to mesh refinement specifically in areas prone to shockwave interactions and received boundary layer adjustments to accurately simulate the flow dynamics. Boundary conditions for the fluid sector, which included an inlet, outlet, symmetry plane, and pressure far field, were essential for facilitating the fluid–thermal–structural integration crucial for an exhaustive analysis. In the solid sector, representing the 2D C/SiC plate, the dimensions were established at 25 mm × 5 mm. Here, the boundary conditions involved applying a heat flux to the C/SiC plate's interface, enabling an exchange of data with the fluid sector and thus ensuring a detailed and accurate modeling process.

Under the impact of a high-energy laser, localized ablation pits are created on the surface of C/SiC composites. The thermal transport within this ablation region, as illustrated in Fig. 9, comprises convective cooling via airflow, laser-induced heating at the material's surface, and heat exchange resulting from material oxidation and sublimation at elevated temperatures. Furthermore, heat radiates into the surrounding environment. The conservation of energy across the ablation boundary is governed by the equation: where q_{in} represents the heat flux into the material, k_{eff} is the effective thermal conductivity, and $\frac{dT}{dn}$ denotes the temperature gradient normal to the surface. The convective heat transfer q_{conv} occurs between the ablation boundary of the material and the ambient environment. The radiant heat loss from the material is denoted by q_{rad} , and q_{ox} and q_{sub} represent the heat released by oxidation and the heat removed by sublimation mass loss, respectively.

$$q_{in} = -k_{eff} \frac{dT}{dn} = q_{laser} - q_{rad} - q_{conv} - q_{sub} - q_{ox} \quad (22)$$

Radiative and convective heat losses are quantified by the following expressions:

$$q_{rad} = \sigma(T^4 - T_0^4) \quad (23)$$

$$q_{conv} = h_c(T - T_0) \quad (24)$$

where σ is the Stefan–Boltzmann constant, T is the surface temperature, and T_0 is the ambient temperature. The convective heat transfer coefficient is h_c .

The heat fluxes related to sublimation and oxidation processes are linearly modeled as:

$$q_{sub} = Q_{sub} \rho_w v_s \quad (25)$$

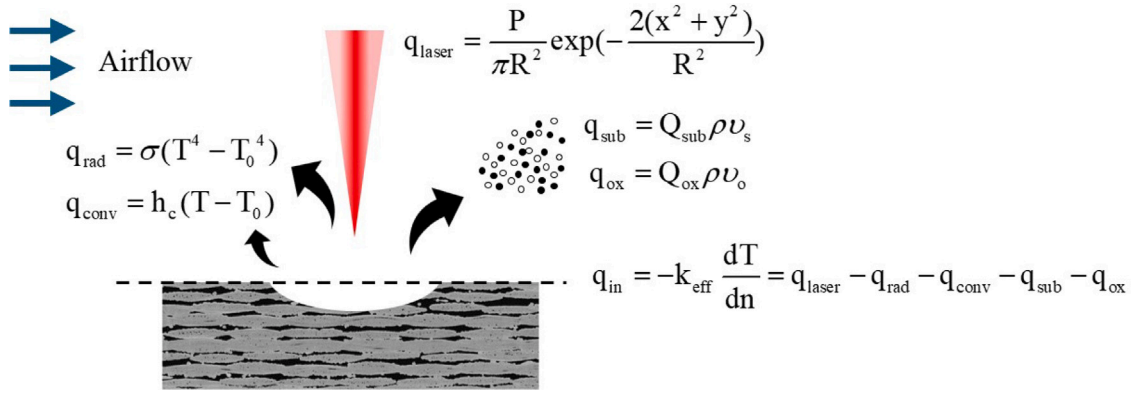


Fig. 9. Energy conservation on ablation surface of C/SiC composite.

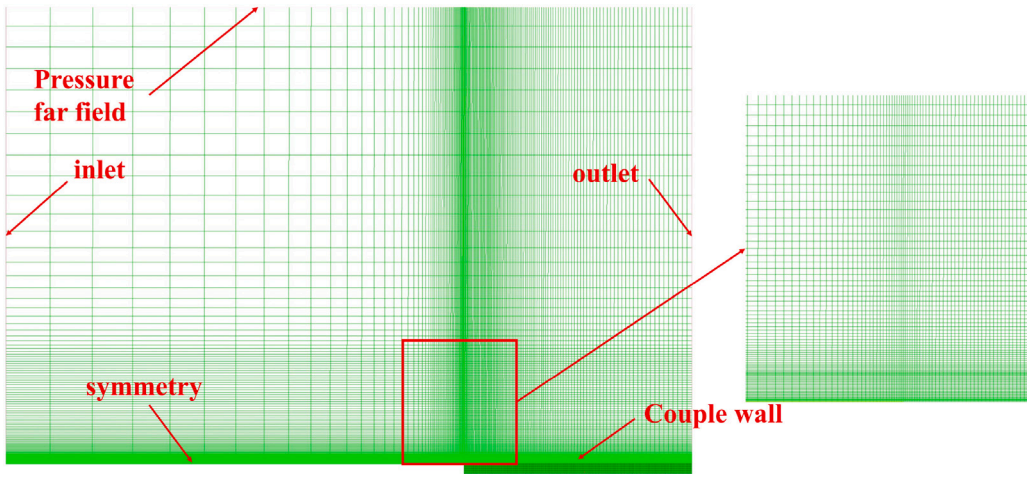


Fig. 10. Grid schematic.

$$q_{\text{ox}} = Q_{\text{ox}} \rho_w v_o \quad (26)$$

where Q_{sub} and Q_{ox} are the reaction heat per unit mass for sublimation and oxidation reactions, respectively.

The model simplifies the effect of a CW laser on the material as a Gaussian-like heat source. The spatial distribution of laser intensity on the material surface is expressed as:

$$q_{\text{laser}} = \frac{P}{\pi R^2} \exp\left(-\frac{2(x^2 + y^2)}{R^2}\right) \quad (27)$$

P is the laser power, and R is the radius of the laser spot. The coordinates x and y are relative to the center of the laser spot.

This section emphasizes the model's unique attributes under low-speed airflow conditions. Previous research indicated that the ablation rate in such scenarios has a minimal dependency on wall shear stress, leading to a simplification of the wall ablation rate formulas as detailed below:

$$v_s = \frac{1}{\rho_w} (\alpha/c_p) \left(p_g^*/p_e(x, t) \right) \exp\left(-\frac{E_{AS}}{RT_w}\right) \quad (28)$$

$$v_o = \frac{1}{\rho_w} \frac{aM_i}{M_{O_2}} p_{O_2w}^n A_i \exp\left(-\frac{E_{i,O}}{RT_w}\right) \quad (29)$$

$$v_{f\perp} = \frac{1}{\rho f} \left(\frac{J_f^0 k_f}{c_f} \right)^{1/2} \left(\frac{6p_d}{\sigma_{f\perp}} \right)^{1/3} \left(\frac{RT_w}{E_{Af}} \right)^{1/2} \exp\left(-\frac{E_{Af}}{RT_w}\right) \quad (30)$$

$$v_{f\parallel} = v_{f\perp} \left(\frac{\sigma_{f\perp}}{\sigma_{f\parallel}} \right)^{1/2} \quad (31)$$

Table 4
Ablation formulation parameters of C/SiC composite [24,26].

Parameters	Value	Unit	Parameters	Value	Unit
ρ_w	2050	kg/m ³	ρ_f	1760	kg/m ³
k_f	42	W/(m K)	k_m	20	W/(m K)
c_f	600	J/(kg K)	c_m	700	J/(kg K)
E_{AS}	8.73×10^4	J/mol	A_C	5.77×10^5	kg/m ³ s
A_{SiC}	10	kg/m ³ s	$E_{C,O}$	1.142×10^5	J/mol
$E_{SiC,O}$	9.46×10^4	J/mol	J_f^0	5.4×10^2	kg/m ³
J_m^0	5×10^5	kg/m ³	$\sigma_{f\parallel}$	2200	MPa
$\sigma_{f\perp}$	220	MPa	σ_m	400	MPa
E_{Af}	5.4×10^4	J/mol	E_{Am}	8.73×10^4	J/mol

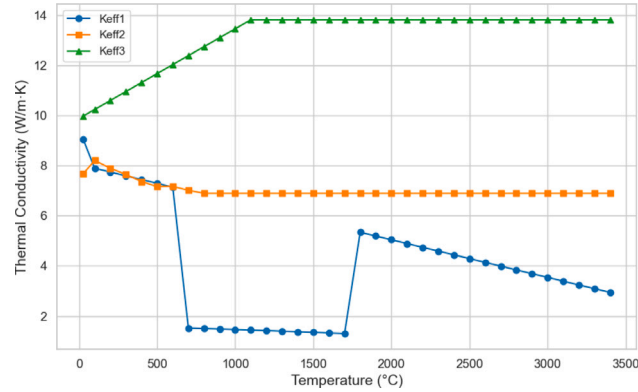


Fig. 11. Thermal conductivity.

$$v_m = \frac{1}{\rho m} \left(\frac{J_m^0 k_m}{c_m} \right)^{1/2} \left(\frac{6p_d(x, t)}{\sigma_m T} \right)^{1/3} \left(\frac{RT_w}{E_{Am}} \right)^{1/2} \exp \left(-\frac{E_{Am}}{2RT_w} \right) \quad (32)$$

ρw denotes the density of C/SiC, and (α/c_p) represents the heat transfer coefficient within the boundary layer. $p_e(x, t)$ signifies the local pressure exerted by the airflow, with p_g^* indicating the initial pressure constant. The term E_{AS} is the activation energy for the sublimation reaction, while T_w refers to the surface temperature. M_i and M_{O_2} are the molecular masses of the reacting substance and oxygen, respectively. The partial oxygen pressure in the airflow is denoted by $p_{O_2, w}^n$. A_i and $E_{i,O}$ represent the pre-exponential factor and activation energy for the oxidation reaction, respectively. The variables f and m pertain to fiber and matrix, respectively, with $v_f \perp$, $v_{f\parallel}$, and v_m indicating their ablation rates. Finally, k and ρ stand for thermal conductivity and density, respectively, with c signifying specific heat capacity.

The relevant material parameters used in this model are shown in Table 4.

3.3.3. Analysis of results

Utilizing the previously developed effective thermal conductivity model, we obtained three distinct thermal conductivities, as depicted in the Fig. 11. Here, Keff1 represents the thermal conductivity calculation method adopted in this study, Keff2 is based on Zhang et al. [8] approach, and Keff3 employs the material property values for C/SiC from the COMSOL software's material library. To evaluate the fitting effectiveness of these methods, we employed two metrics: the coefficient of determination (r^2) and the root mean squared error (RMSE). The calculation formulas are as follows:

$$r^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (33)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (34)$$

where y_i is the observed value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the mean of the observed values.

Fig. 12 depicts the temperature variations, showcasing the efficacy of our fitting method. By incorporating the ablation model previously delineated, we investigated the variation of effective thermal conductivity with temperature across different power densities and scrutinized the corresponding fitting outcomes. At a laser power density of 330 W/cm², our method yielded temperature predictions that were more congruent with experimental data compared to alternative approaches. This alignment is evidenced by the improved calibration of temperature rise rates and the reduced temperature discrepancies observed between the

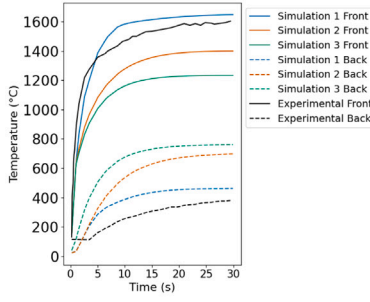
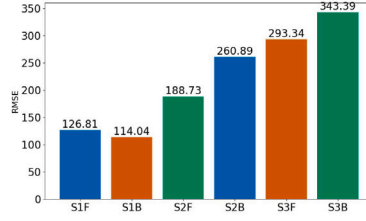
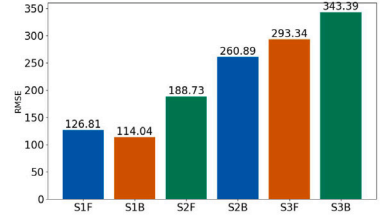
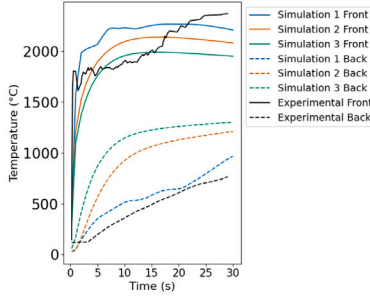
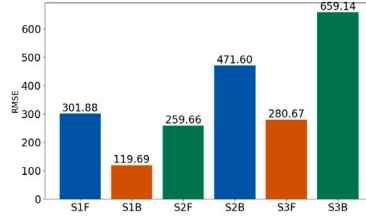
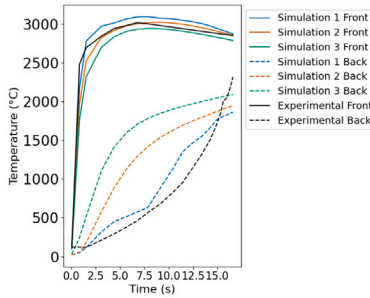
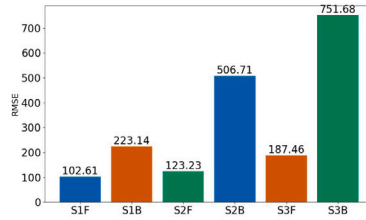
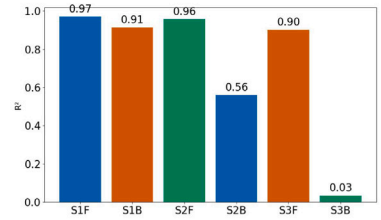
(a) Temperature 330W/cm²(b) RMSE 330W/cm²(c) R²330W/cm²(d) Temperature 650W/cm²(e) RMSE 650W/cm²(f) R²650W/cm²(g) Temperature 1420W/cm²(h) RMSE 1420W/cm²(i) R²1420W/cm²

Fig. 12. Comparison plot of fitting effect.

front and rear surfaces, substantiated by the RMSE and r^2 metrics. Notably, the fitting accuracy on the front surface was significantly enhanced. However, the persistent temperature disparity at the rear surface indicates potential oversight of factors such as product thermal resistance or secondary effects, necessitating further exploration.

With a laser power density set at 650 W/cm², our advanced thermal conductivity model, which accounts for by-products, successfully met two principal objectives. Firstly, it sustained stable ablation rates on the rear surface, aligning with experimental observations. Secondly, it captured the front surface's gradual temperature increase to approximately 1700 °C, marked by significant fluctuations indicative of by-product influence. While the model's portrayal of the fluctuation's duration and magnitude was marginally less accurate than the experimental findings – resulting in a slight reduction in the front surface's RMSE accuracy at this stage – the ultimate temperature prediction was exceptionally precise. Notably, the rear surface simulations outperformed other models in terms of the coefficient of determination and RMSE, highlighting the importance of further analyzing the heat transfer dynamics on the front surface during the temperature rise.

At a laser power density of 1420 W/cm², we noted ablation through-holes after 17 s, using this as our experimental reference for ablation temperature data up to that point. Post-peak, temperatures fell due to diminished ablation and airflow effects. Beyond 2800 °C, both surfaces exhibited outstanding RMSE and r^2 values. Even amidst the rapid heating and brief reaction times characteristic of high-power laser irradiation, integrating reaction processes into our temperature profile analysis markedly improved our temperature prediction accuracy. Notably, by implementing a temperature-dependent thermal conductivity model, the rear surface's temperature progression effectively illustrated an accelerated rate of temperature increase, a nuance more precisely

reflecting the experimental data. This distinct trend of temperature rise rate on the rear surface, captured by our model, sets it apart from other models.

In conclusion, our model exhibited superior performance relative to others across a range of irradiation conditions, with notable effectiveness at 1420 W/cm^2 . However, the observed temperature differences on the rear surface at 330 W/cm^2 and the heat transfer mechanisms that contribute to the temperature fluctuations on the front surface and the gradual temperature increase at 650 W/cm^2 require further investigation to refine our model's predictive accuracy and reliability.

4. Conclusion

In this research, we conducted laser ablation experiments in a wind tunnel under airflow conditions at 0.6 Ma , with power densities ranging from 330 to 1420 W/cm^2 . Our objectives were to examine ablation depth, morphology, and temperature fluctuations on both the front and rear surfaces of C/SiC composites. The key findings are:

1. Significant temperature differences were observed between the front and back surfaces during heat transfer. The front surface showed a gradual temperature increase with fluctuations around 1700°C , while the back surface exhibited a rapid rise in temperature with increased power density.
2. Our heat transfer model, which incorporates ablation by-products, demonstrated improved accuracy in high-temperature environments. This was validated by comparing RMSE and r^2 values against experimental results, particularly at higher temperatures.
3. A microscale model based on porosity was developed to determine the thermal properties of $\text{SiO}_2(s)$, accurately predicting behaviors up to 1723°C . Beyond this temperature, liquid $\text{SiO}_2(l)$ also influenced heat transfer, even under 0.6 Ma conditions.
4. Persistent temperature discrepancies and fluctuations on the rear surface at lower power densities might be due to the thermal resistance of by-products or the presence of $\text{SiO}_2(s)$. These factors require further investigation to fully understand their impact on heat transfer.

In conclusion, our model exhibited superior performance relative to others across a range of irradiation conditions, with notable effectiveness at 1420 W/cm^2 . However, the observed temperature differences on the rear surface at 330 W/cm^2 and the heat transfer mechanisms that contribute to the temperature fluctuations on the front surface and the gradual temperature increase at 650 W/cm^2 require further investigation to refine our model's predictive accuracy and reliability.

Future work will focus on:

- Investigating the applicability of our dynamic thermal conductivity model to a broader range of composite materials, including those with different compositions and microstructures.
- Conducting experiments at higher laser power densities to assess the model's performance under more extreme thermal conditions.
- Studying the long-term effects of repeated thermal cycling on material stability and performance, providing insights into the durability and reliability of thermal protection systems over extended periods of use.
- Exploring the impact of high Mach number airflow conditions on the ablation behavior and thermal conductivity of C/SiC composites, addressing the limitations identified in the current study.

These future research directions aim to address the identified gaps and improve the robustness and applicability of our thermal conductivity model across various conditions and materials. The insights gained from this study lay a solid foundation for the development of advanced thermal protection systems, enhancing the safety and performance of aerospace applications under extreme conditions.

CRediT authorship contribution statement

Lei Wang: Methodology, Software, Writing – review & editing. **Chuntong Liu:** Visualization, Investigation. **Zhili Zhang:** Supervision, Conceptualization. **Ran Zhang:** Data curation, Writing – original draft.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ran Zhang reports equipment, drugs, or supplies was provided by State Key Laboratory of Laser Interaction with Matter, Northwest Institute of Nuclear Technology. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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